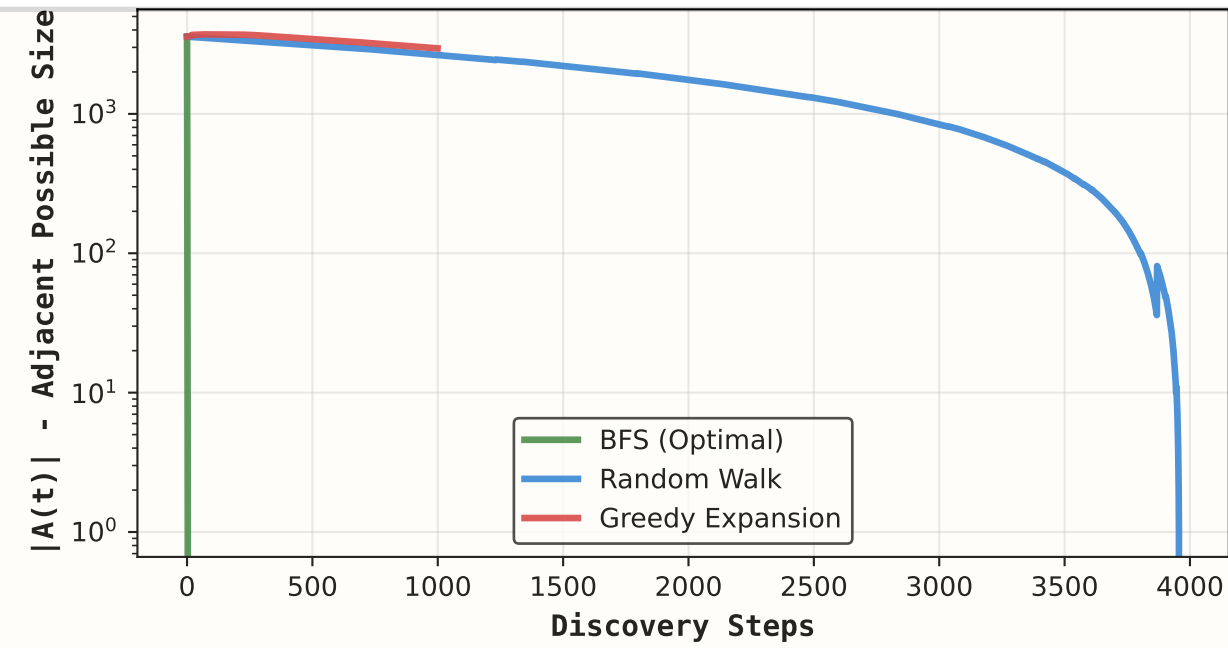
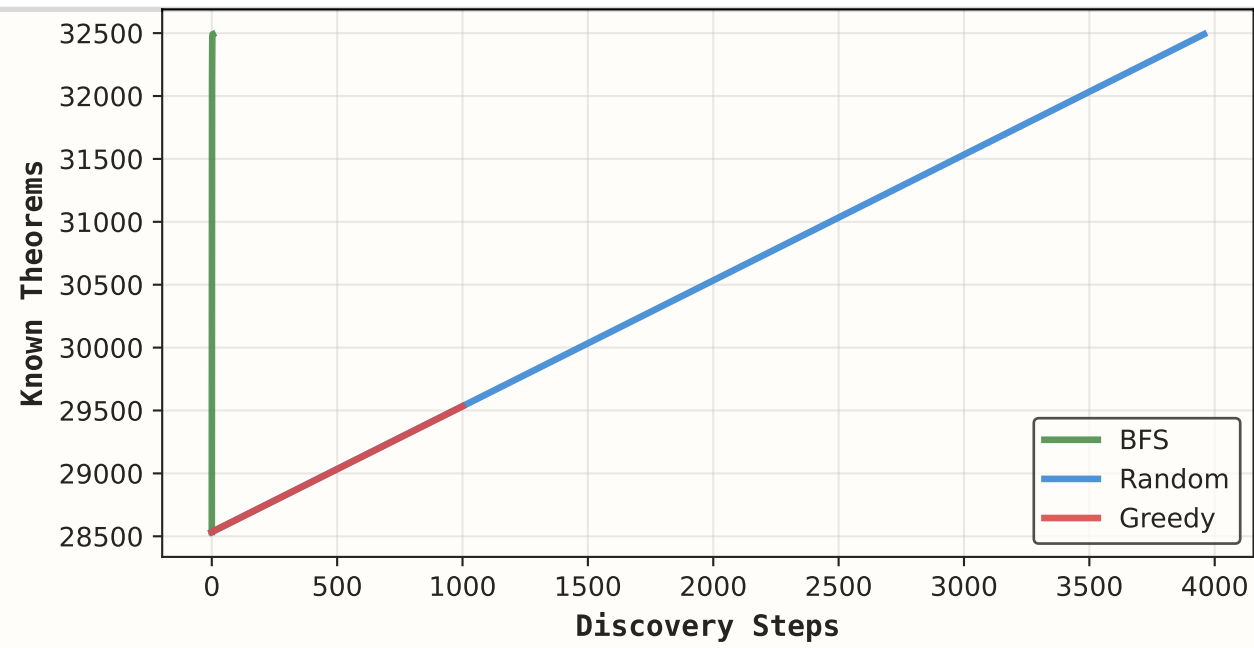


Mathematical Discovery Dynamics: The Adjacent Possible in Formal Theorem Proving

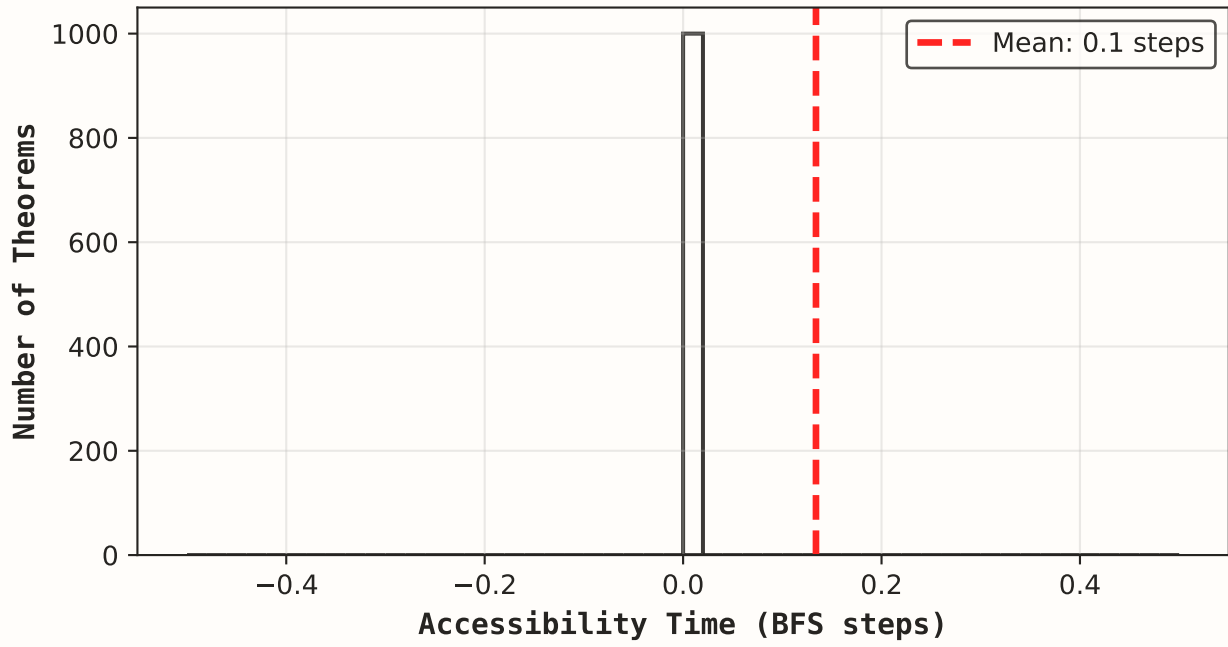
A. Evolution of Possibility Space



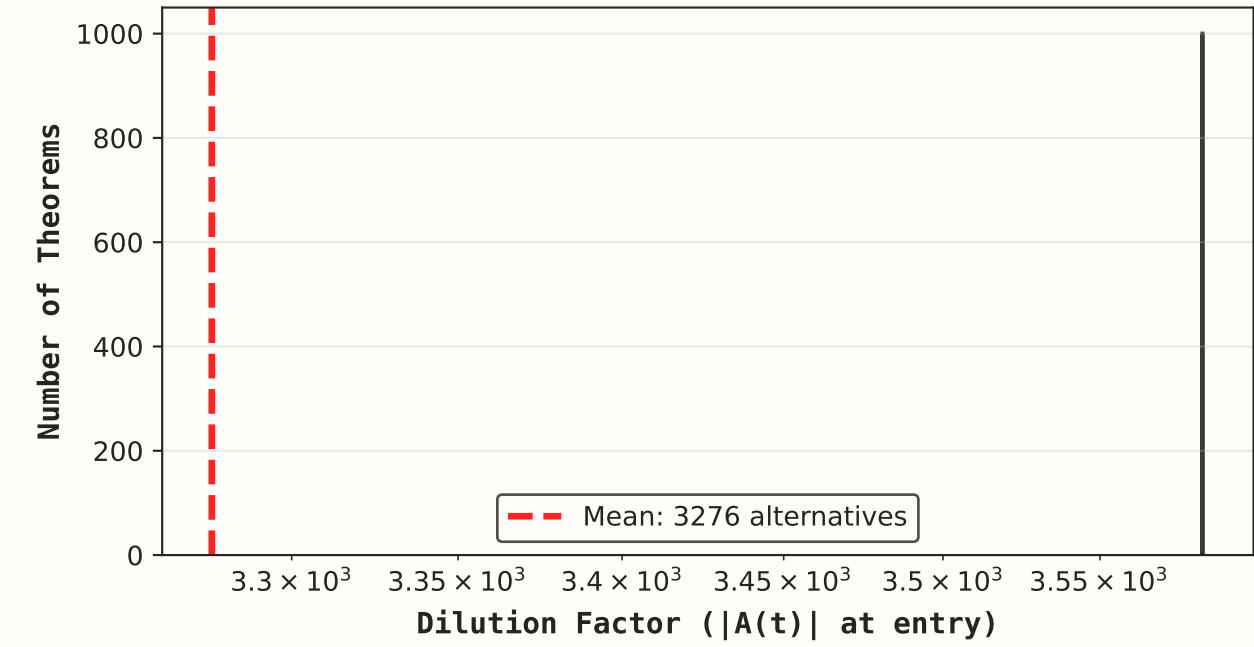
B. Knowledge Accumulation



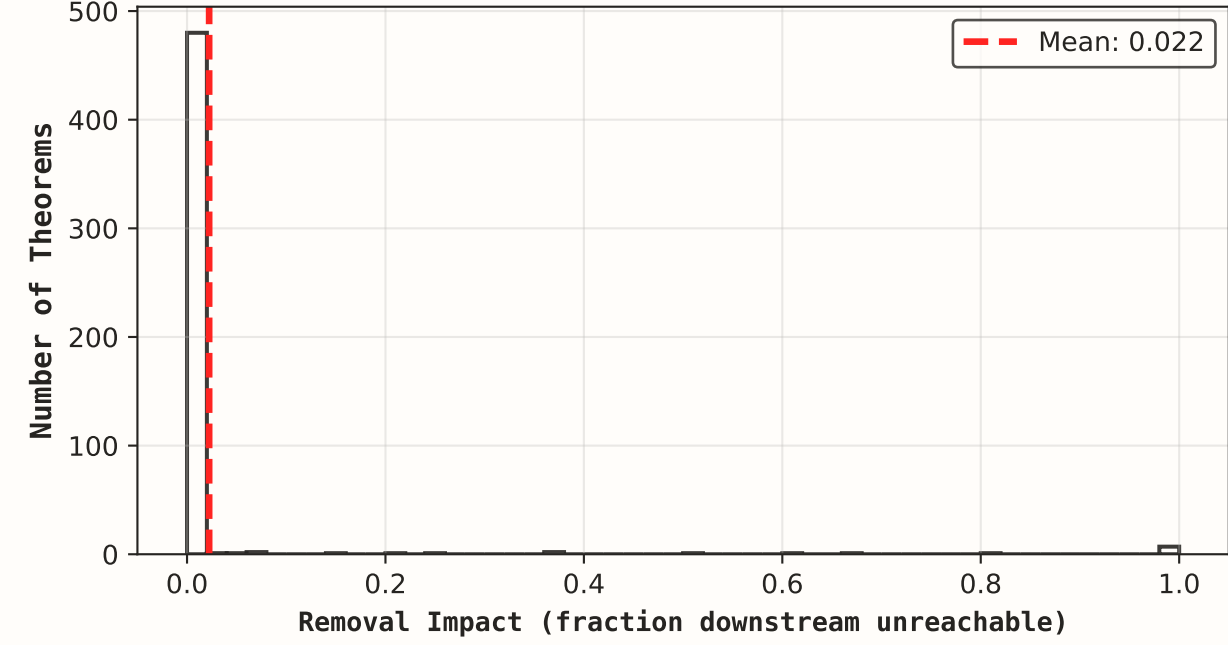
C. When Theorems Become Provable



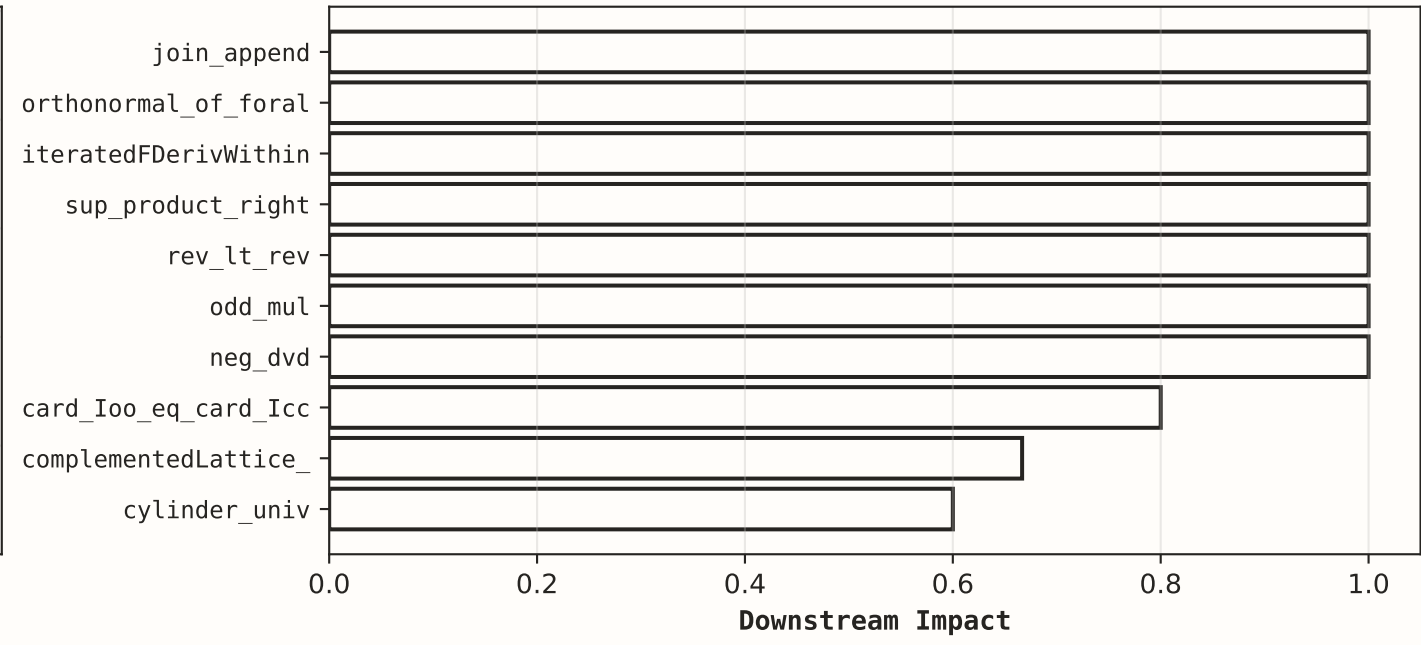
D. "Hiding in Plain Sight"



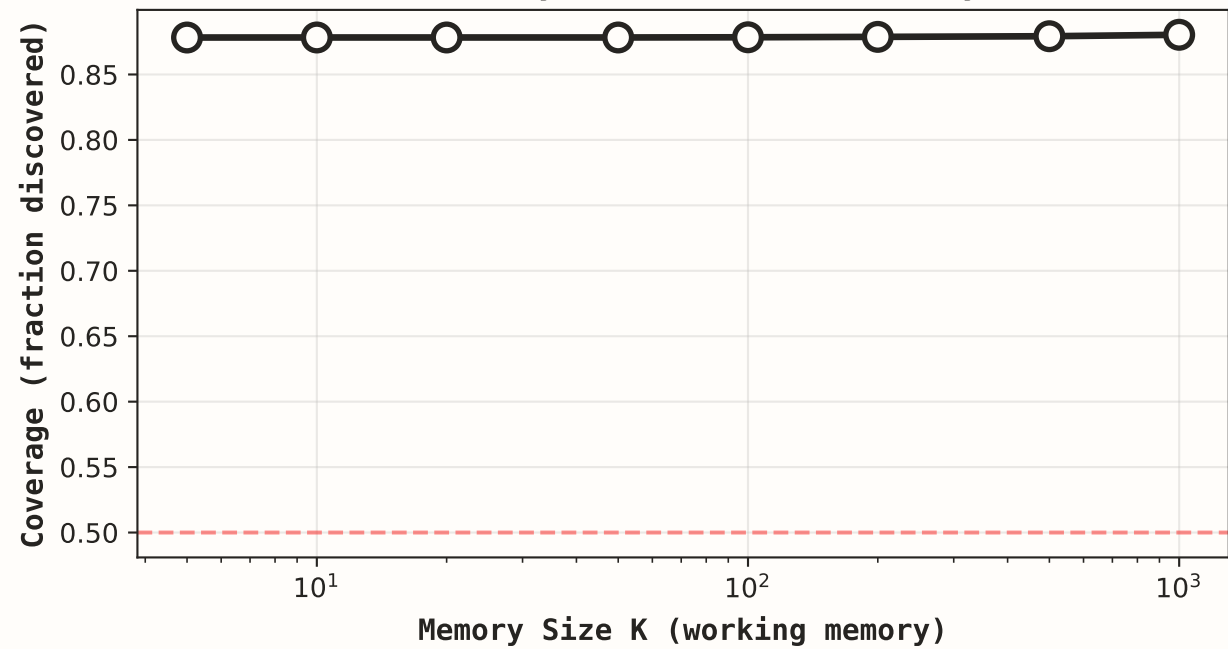
E. Gateway Theorem Identification



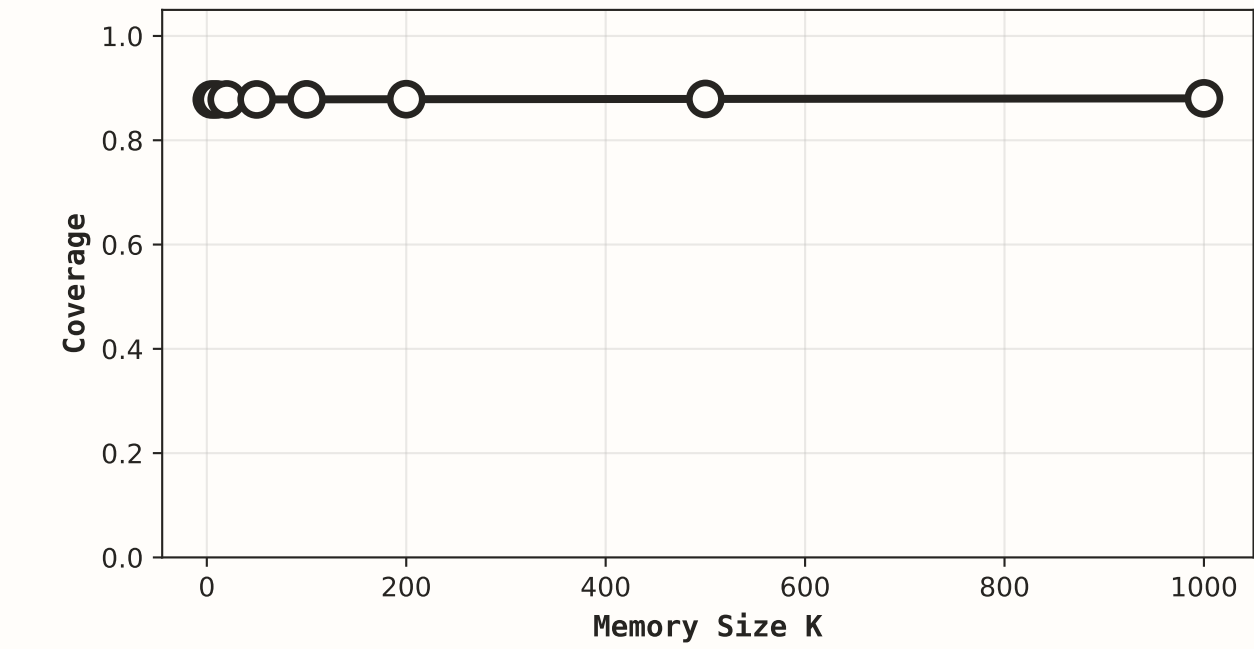
F. Top 10 Gateway Theorems



G. Memory-Constrained Discovery



H. Phase Transition Search



FINDING: Explosive Growth Phase

Adjacent possible $|A(t)|$ shows:

- BFS: Rapid initial explosion, then plateau as graph exhausted
- Random: Slower growth, higher peak (explores broadly before depth)
- Greedy: Optimizes expansion but similar to BFS

Coverage achieved:

- BFS: 100.0%
- Random: 100.0%
- Greedy: 90.9%

INTERPRETATION:

Mathematical discovery is NOT uniform exploration. Early discoveries unlock explosive growth in possibilities, then possibility space contracts as easier results are exhausted.

Greedy $\approx$ BFS suggests humans naturally optimize expansion, not randomly walk.

FINDING: Dilution Effect

Accessibility time: When theorem first becomes provable (all prerequisites met)

Mean accessibility: 0.1 steps
Max accessibility: 4 steps

Dilution factor: How many alternatives exist when theorem enters $A(t)$

Mean dilution: 3276 options
Max dilution: 3,583 options!

INTERPRETATION:

Theorems entering $A(t)$ when $|A(t)|$ is large are "hidden" among many options. Discovery difficulty $\neq$ just depth, but also how diluted the theorem is.

High-dilution theorems require either:

- Strategic search (not random)
- Explicit value recognition
- Lucky exploration

FINDING: Sparse Gateways

Removal impact: Fraction of downstream theorems unreachable if theorem removed

Mean impact: 0.022
Max impact: 1.000

Top gateway theorem:

join_append
Impact: 1.000

INTERPRETATION:

Mathematical knowledge does NOT have bow-tie structure with critical narrow gateways. Most theorems have low removal impact.

This suggests robust interconnection: many alternative paths to any result.

Discovery order relatively unimportant for ultimate coverage - multiple routes exist to most theorems.

FINDING: Memory Threshold

Bounded working memory limits what can be discovered. Theorem enters $A(t)$ only if ALL prerequisites in last K discovered theorems.

Coverage results:

- K=10: 87.8%
- K=100: 87.8%
- K=500: 87.9%

INTERPRETATION:

Smooth growth (not sharp transition) suggests continuous difficulty spectrum.

Implications for bounded agents:

- $K < 50$: Severely limited discovery
- $K > 200$: Most mathematics accessible
- No single threshold complexity

Human mathematicians use external memory (papers, notes) to exceed biological working memory limits!